

GROUP

Homomorphism

$$\varphi: G_1 \longrightarrow G_2$$

Definition: Suppose G and H are groups. A map

$$\varphi: G \rightarrow H$$

is called a homomorphism if

$$\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y) \quad \forall x, y \in G$$

example: $\varphi: \mathbb{Z} \rightarrow G$, G any group

$$\varphi(n) = g^n \text{ where } g \in G \text{ is fixed.}$$

$$\varphi(m+n) = g^{m+n} = g^m \cdot g^n = \varphi(m) \varphi(n)$$

ex.: $\varphi: GL_2(\mathbb{R}) \rightarrow \mathbb{R}^*$, $\varphi(A) = \det(A)$

$$\varphi(A \cdot B) = \det(A \cdot B) = \det(A) \cdot \det(B) = \varphi(A) \cdot \varphi(B).$$

ex.: $\varphi: (\mathbb{R}, +) \rightarrow \mathbb{T} = S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$

$$\varphi(x) = e^{ix}, \quad \varphi(x+y) = e^{i(x+y)} = e^{ix} e^{iy} = \varphi(x) \cdot \varphi(y)$$

proposition: Let $\varphi: G_1 \rightarrow G_2$ be a homomorphism then

$$\textcircled{a} \text{ If } e_1 \in G_1 \text{ then } e_2 = \varphi(e_1) \in G_2$$

proof: Let $x \in G_1 \Rightarrow e_1 x = x \Rightarrow \varphi(e_1 x) = \varphi(x)$

$$\Rightarrow \varphi(e_1) \varphi(x) = \varphi(x). \text{ Now left mul with } [\varphi(x)]^{-1}$$

$$\Rightarrow \varphi(e_1) = \varphi(x) \cdot [\varphi(x)]^{-1} = e_2 \quad \square$$

proposition: Let $\varphi: G_1 \rightarrow G_2$ be a homomorphism then

$$\textcircled{2} \quad \forall x \in G_1 \quad \varphi(x^{-1}) = [\varphi(x)]^{-1}$$

proof: $e_1 = x \cdot x^{-1}$

$$\begin{aligned} e_2 &= \varphi(e_1) = \varphi(x) \cdot \varphi(x^{-1}) \quad \text{Left mul } [\varphi(x)]^{-1} \\ \Rightarrow [\varphi(x)]^{-1} &= \varphi(x^{-1}) \end{aligned}$$

proposition: Let $\varphi: G_1 \rightarrow G_2$ be a homomorphism then

$$\textcircled{4} \quad \text{If } H_2 \leq G_2 \text{ then } \varphi^{-1}(H_2) \leq G_1$$

$$\textcircled{4} \quad (H_2 \trianglelefteq G_2) \quad (\varphi^{-1}(H_2) \trianglelefteq G_1)$$

proof: Recall subgroup test: $H \leq G \Leftrightarrow (\forall x, y \in H \Rightarrow xy^{-1} \in H)$

$$\text{Suppose } x, y \in \varphi^{-1}(H_2). \Rightarrow \varphi(x), \varphi(y) \in H_2 \leq G_2$$

$$\Rightarrow \varphi(x)[\varphi(y)]^{-1} \in H_2 \Rightarrow \varphi(xy^{-1}) \in H_2$$

$$\Rightarrow xy^{-1} \in \varphi^{-1}(H_2) \leq G_1$$

$$\text{Suppose } H_2 \trianglelefteq G_2 \quad \boxed{N \trianglelefteq G \Leftrightarrow xnx^{-1} \in N \quad \forall x \in G, n \in N}$$

$$\begin{aligned} \text{Take } n \in \varphi^{-1}(H_2), x \in G_1 &\Rightarrow \varphi(xnx^{-1}) = \varphi(x)\varphi(n)\varphi(x^{-1}) \\ &\in H_2 \text{ b/c } H_2 \trianglelefteq G_2 \end{aligned}$$

$$\Rightarrow xnx^{-1} \in \varphi^{-1}(H_2) \rightarrow \varphi^{-1}(H_2) \trianglelefteq G$$

proposition: Let $\varphi: G_1 \rightarrow G_2$ be a homomorphism then

$$\textcircled{1} \quad \text{If } e_1 \in G_1 \Rightarrow e_2 = \varphi(e_1) \in G_2$$

$$\textcircled{2} \quad \forall x \in G_1 \quad \varphi(x^{-1}) = [\varphi(x)]^{-1}$$

$$\textcircled{3} \quad \text{If } H_1 \leq G_1 \Rightarrow \varphi(H_1) \leq G_2$$

$$\textcircled{4} \quad \text{If } H_2 \leq G_2 \Rightarrow \varphi^{-1}(H_2) \leq G_1$$

$$H_2 \trianglelefteq G_2 \Rightarrow \varphi^{-1}(H_2) \trianglelefteq G_1$$

Definition: The kernel of φ :

$$\ker \varphi = \{x \in G_1 \mid \varphi(x) = e_2\} = \varphi^{-1}(e_2)$$

$$\underline{\text{ex.}}: \varphi: \mathbb{Z} \rightarrow \mathbb{Z}_5, \varphi(n) = [n]$$

$$\begin{aligned} \ker \varphi &= \{n \in \mathbb{Z} \mid \varphi(n) = [0]\} = \{n \in \mathbb{Z} \mid [n] = [0]\} \\ &= \{n \in \mathbb{Z} \mid n \equiv 0 \pmod{5}\} = 5\mathbb{Z}. \end{aligned}$$

$$\underline{\text{ex.}}: \varphi: \mathbb{R} \rightarrow \mathbb{C}^*, \varphi(x) = e^{2\pi i x}$$

$$\ker \varphi = \{x \in \mathbb{R} \mid e^{2\pi i x} = 1\} = \mathbb{Z}$$

$$\stackrel{\text{b/c}}{\Rightarrow} \cos(2\pi x) + i \sin(2\pi x) = 1 \Rightarrow x \in \mathbb{Z}$$

proposition: $\ker \varphi \trianglelefteq G_1$ ($\varphi: G_1 \rightarrow G_2$)

proof: subgroup ($H \leq G \Leftrightarrow \forall x, y \in H \Rightarrow xy^{-1} \in H$)

Suppose $x, y \in \ker \varphi$. Consider xy^{-1} and

$$\varphi(xy^{-1}) = \varphi(x) \cdot \varphi(y^{-1}) = \varphi(x) \cdot [\varphi(y)]^{-1} = e \cdot e^{-1} = e$$

$\Rightarrow xy^{-1} \in \ker \varphi$ and thus $\ker \varphi \leq G_1$

normal: ($N \trianglelefteq G \Leftrightarrow \forall n \in N, x \in G \Rightarrow xnx^{-1} \in N$)

Suppose $x \in G_1, n \in \ker \varphi$

$$\begin{aligned} \Rightarrow \varphi(xnx^{-1}) &= \varphi(x) \cdot \varphi(n) [\varphi(x)]^{-1} \\ &= \varphi(x) \cdot [\varphi(x)]^{-1} = e_2 \end{aligned}$$

$\Rightarrow xnx^{-1} \in \ker \varphi \Rightarrow \ker \varphi \trianglelefteq G_1$

First Isomorphism Theorem

If $\varphi: G \rightarrow H$ is a homomorphism and $\pi: G \rightarrow G/\ker \varphi$

then $\exists!$ isomorphism $\psi: G/\ker \varphi \rightarrow \text{im}(\varphi) \leq H$ such that $\psi \circ \pi = \varphi$

proof: Let $\psi: G/\ker \varphi \rightarrow H, \psi(x \ker \varphi) = \varphi(x) \in \text{im}(\varphi) \leq H$.

Well defined: Suppose $x \ker \varphi = y \ker \varphi$

$$\Rightarrow xy^{-1} \in \ker \varphi \Rightarrow \varphi(xy^{-1}) = e_H$$

$$\begin{aligned} \Rightarrow \varphi(x) \cdot [\varphi(y)]^{-1} &= e_H \Rightarrow \varphi(x) = \varphi(y) \\ &\quad \parallel \quad \parallel \\ &\quad \psi(x \ker \varphi) \quad \psi(y \ker \varphi) \end{aligned}$$

Homomorphism:

$$\begin{aligned} \psi((x \ker \varphi)(y \ker \varphi)) &= \psi(xy \ker \varphi) = \varphi(xy) = \varphi(x) \cdot \varphi(y) \\ &= \psi(x \ker \varphi) \cdot \psi(y \ker \varphi) \end{aligned}$$

Injective: Suppose $x \ker \varphi \in \ker \psi$

$$\begin{aligned} \Rightarrow \psi(x \ker \varphi) &= e \\ &\quad \parallel \\ &\quad \varphi(x) \end{aligned}$$

$$x \in \ker \varphi \Rightarrow x \ker \varphi = e \ker \varphi = \ker \varphi$$

$$\Rightarrow \ker \psi = \{ \ker \varphi \}.$$

surjective: $\psi(x \ker \varphi) = \varphi(x)$

Suppose $y \in \text{im} \varphi \Rightarrow \exists x \in G$ such that $\varphi(x) = y$.

note: $\psi(x \ker \varphi) = \varphi(x) = y$

uniqueness: $\psi(x \ker \varphi) = \varphi(x)$ ($\pi(x) = x \ker \varphi$)

Suppose $\bar{\varphi}: G/\ker \varphi \xrightarrow{\sim} \text{im } \varphi$ such that $\bar{\varphi}\pi = \varphi$.

Take $x \ker \varphi \in G/\ker \varphi$:

$$\bar{\varphi}(x \ker \varphi) = \bar{\varphi}(\pi(x)) = \varphi(x) = \varphi(\pi(x)) = \varphi(x \ker \varphi)$$

Suppose G is a group.

$$\text{Aut}(G) := \{ \varphi: G \rightarrow G \mid \varphi \text{ is an iso} \}$$

$$\text{Inn}(G) := \{ \varphi: G \rightarrow G \mid \varphi(x) = g x g^{-1} \}$$

$$Z(G) := \{ g \in G \mid g x = x g \quad \forall x \in G \}$$

claim: $\text{Aut}(G)$ form a group with composition

sketch: ① identity $i: G \rightarrow G$, $i(g) = g$

② inverses: if $\varphi: G \rightarrow G$ is an iso, so is φ^{-1}

③ associative: follows from the associativity of composition of functions.

④ closure: also from composing automorphisms

$\text{Inn}(G) \leq \text{Aut}(G)$:

subgroup test: $H \leq G \iff \forall x, y \in H \Rightarrow xy^{-1} \in H$

$$i_x, i_y \in \text{Inn}(G)$$

note: $(i_y)^{-1} = i_{y^{-1}}$ because $i_y \circ i_{y^{-1}}(g)$
 $= y(y^{-1} g y) y^{-1} = g$

note: $i_x \circ i_{y^{-1}}(g) = x(y^{-1} g y) x^{-1} = (xy^{-1}) g (xy^{-1})^{-1}$
 $= i_{xy^{-1}}(g)$

$$\Rightarrow i_x \circ i_{y^{-1}} = i_{xy^{-1}} \in \text{Inn}(G)$$

claim: $G/Z(G) \cong \text{Inn}(G)$

proof: $\varphi: G \rightarrow \text{Inn}(G) \leq \text{Aut}(G)$

$$\varphi(g) = i_g \quad (i_g(x) = g x g^{-1})$$

Homomorphism: $\varphi(gh)(x) = i_{gh}(x) = (gh)x(gh)^{-1}$
 $= g(hxh^{-1})g^{-1} = i_g i_h(x)$
 $= \varphi(g)\varphi(h)(x) \quad \checkmark$

Surjective: $\varphi(g) = i_g$, take g to get i_g . ✓

$$\ker \varphi = \{g \in G \mid \varphi(g) = i_g = \text{id}\}$$

$$i_g(x) = \text{id}(x) = x \quad \forall x \in G$$

$$\Leftrightarrow gxg^{-1} = x \quad \Leftrightarrow gx = xg \quad \forall x \in G$$

$$\Rightarrow \ker \varphi = Z(G)$$

$$\Rightarrow \frac{G}{\ker \varphi} = \frac{G}{Z(G)} \cong \text{Inn}(G)$$

□